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SECTOR AT A GLANCE 03

Irrigation Water Management in India

Water in India

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IRRIGATION WATER MANAGEMENT IN INDIA

WATER IN INDIA

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

Suggested Citation:

NITI Aayog (2023). Irrigation Water Management In India. NITI Aayog Sector-at-a-Glance Series Series – Water in India ,2023 -Water -SAG-03, <link>

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Overview

Irrigation water management refers to the complete lifecycle of processes and systems - ranging from water source creation, allocation, conveyance, on-farm application and drainage - all aimed towards enhancing water use efficiency in agriculture and is critically essential for India's food production goals, economic growth, and overall development.¹ Adopting effective IWM practices can lead to improved agricultural productivity while simultaneously ensuring the sustainability of groundwater and surface water resources.

The importance of irrigation for India has been long recognized and investments have been made to create a large network of dams, canals, and other irrigation infrastructure. India's irrigation potential is 140 million hectares. 52% of India's gross cropped area is covered by irrigation.² Nevertheless, around 90% of agriculture in India relies on surface irrigation, resulting in a water-use efficiency of approximately 30-40%, significantly lower than that observed in more developed economies by about 35-40%.

There exist significant opportunities to increase the adoption of participatory irrigation management (PIM), improve the management of irrigation assets through the collection of appropriate fees, expand the use of micro-irrigation systems, drive behaviour change programs, backed with appropriate incentives within farming communities that drive more efficient water use in farming and also a shift away from water-guzzling crops towards more locally suited crops such as millets, and finally more scientific and data-driven approaches to improve the efficiency of irrigation systems.

Key government programs, such as the Pradhan Mantri Krishi Sinchayee Yojana (PMKSY), focus on this by enhancing water use efficiency, improving PIM, and ensuring sustainable irrigation infrastructure growth and maintenance.³ Similarly, several state- and local-level policies and initiatives such as Gujarat's Sujalam Sufalam Yojana and micro-irrigation programs in Andhra Pradesh and Maharashtra among others seek to improve various aspects of irrigation water management in India.⁴

This note provides a brief overview of irrigation water management in India and associated sub-topics.

Figure 1: Key irrigation statistics

75 million hectares

Net irrigated area in India

217.61 BCM

Current annual groundwater extraction for irrigation

52%

Gross cropped area under irrigation coverage

14.4 million hectares

Total area covered under micro-irrigation

Source: Ministry of Agriculture and Farmers Welfare, Agriculture Statistics at a Glance, 2022; Lok Sabha Secretariat, Standing Committee on Water Resources, 2023; NITI Aayog, A New Paradigm for Indian Agriculture, 2022

Note: BCM refers to billion cubic metres.

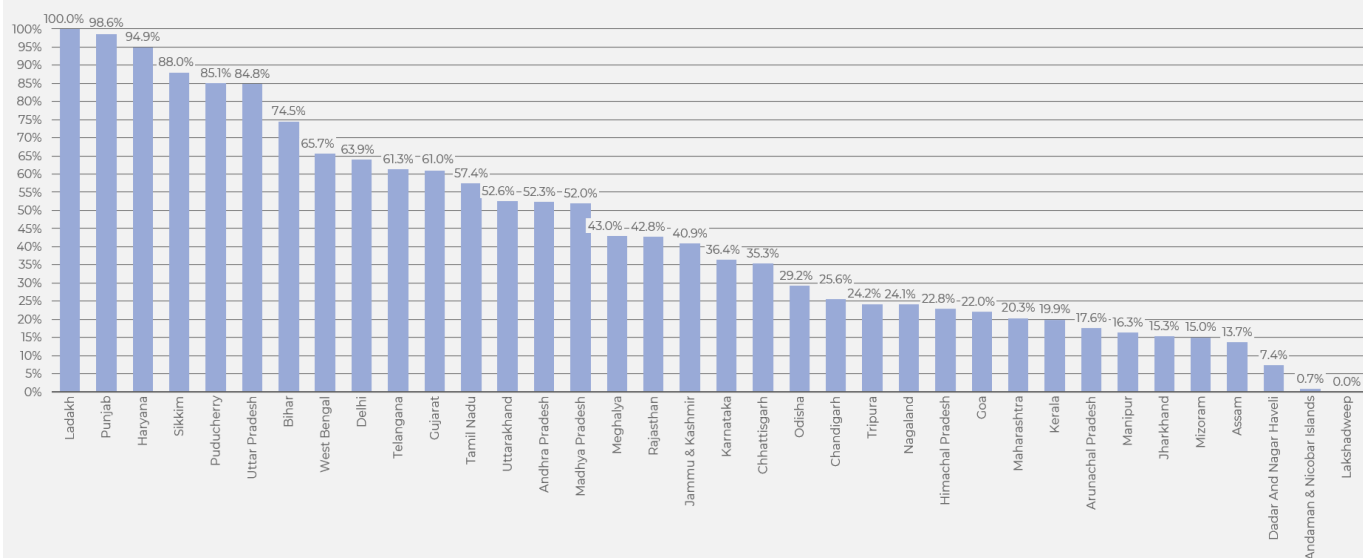
¹ UN Water, SDG 6 Progress, 2019; Lok Sabha Secretariat, Standing Committee on Water Resources, 2023

² NITI Aayog, A New Paradigm for Indian Agriculture, 2023

³ Ministry of Jal Shakti, Pradhan Mantri Krishi Sinchayee Yojana (PMKSY), 2022

⁴ ICAR, Efficiency of Micro-Irrigation in economizing water use in India, 2019

Figure 2: State-wise irrigation coverage of area under principal crops (2019-20)



Source: Ministry of Agriculture and Farmers Welfare, Agricultural Statistics at a Glance, 2022

Key trends

This section dives deeper into India's irrigation water management landscape, outlining statistics and trends on:

1. Participatory Irrigation Management (PIM)
2. Command Area Development (CAD)
3. Water-Use Efficiency (WUE)
4. Water Pricing
5. Equity and Justice
6. Governance and Capacity Building

Participatory Irrigation Management (PIM)

PIM, which involves engaging communities in irrigation management, is crucial for effective water resource management for agriculture in India.

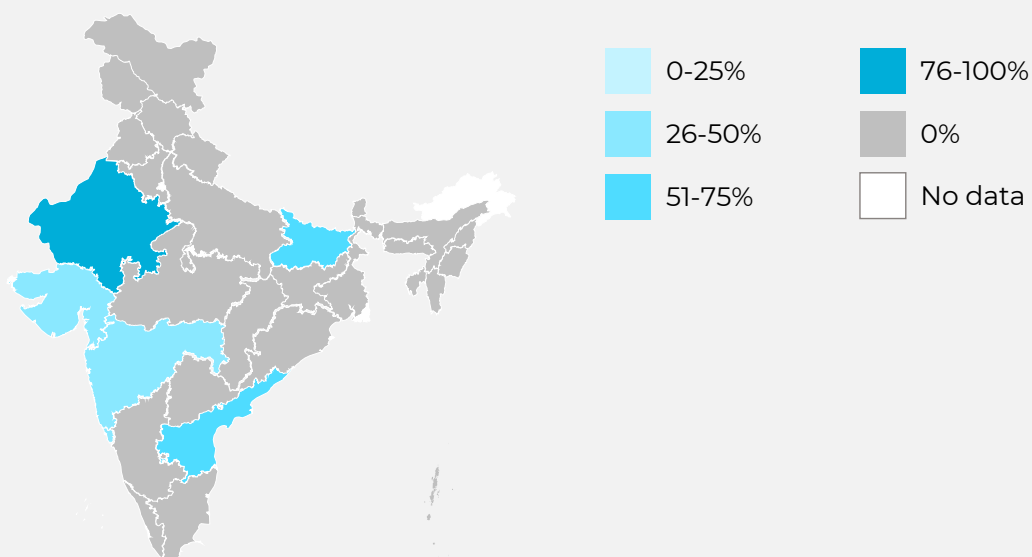
Most states have legal frameworks for PIM in India:

80% of the reporting states and UTs have established a legal framework to facilitate PIM by involving Water User Associations (WUAs).⁵

Involvement of WUAs in O&M of irrigation facilities:

Despite established legal frameworks, only 36% of the reporting states and UTs have active involvement of WUAs in the operations and maintenance of irrigation facilities for more than 50% of their total irrigated area. The situation is notably worse in the states of Bihar and Uttar Pradesh, with less than 5% of the irrigated area with the involvement of WUAs.⁶

Figure 3: Percentage of irrigation service fee retained by WUAs as compared to the fee collected by WUAs during the FY (in per cent, FY 17-18)



Source: NITI Aayog, Composite Water Management Index 2.0, 2019

⁵ NITI Aayog, Composite Water Management Index 2.0, 2019

⁶ NITI Aayog, Composite Water Management Index 2.0, 2019

Command Area Development (CAD)

CAD refers to mechanisms required to effectively deliver water to the farmer's fields and maximize benefits derived from irrigation projects. It includes activities such as lining canals, levelling & shaping land in the command area, drainage management, and engaging with farmers to promote appropriate cropping patterns and practices to efficiently use irrigation water. CAD has been a national priority of the government of India since the 1970s to bridge the gap between irrigation potential created (IPC) and irrigation potential utilized (IPU) and improve agricultural productivity.

India is close to achieving its ultimate irrigation potential (UIP):

The total UIP in India is 140 million hectares (Mha), and 80% of that, 112 Mha has already been created (IPC). Projects sanctioned through PMKSY-Accelerated Irrigation Benefit Programme (AIBP) are developing the remainder.⁷

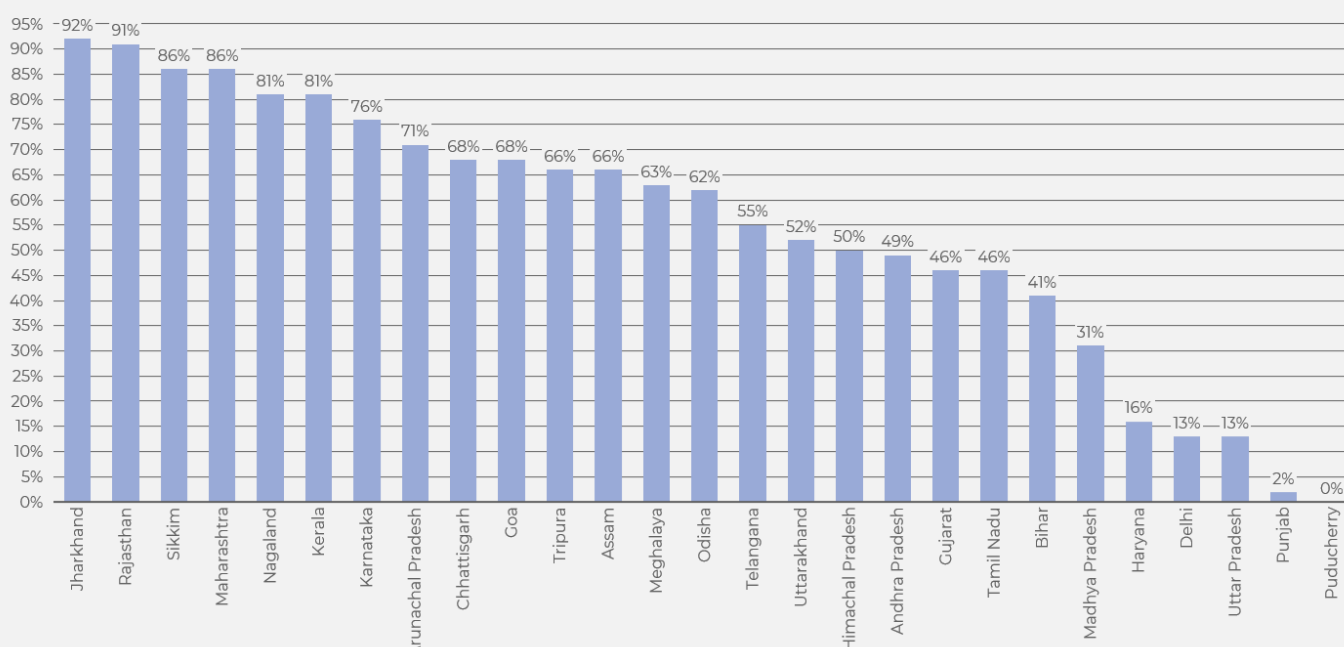
There are gaps in the utilisation of India's irrigation potential:

Out of 112 million hectares (Mha) of total IPC, 19 Mha (16%) gap is left in irrigation potential utilised (IPU), implying under-utilisation of irrigation infrastructure.

With several positive outcomes to show, CAD programs now need to focus on sustainability:

Previous evaluations of the CAD program point to 97% of the target farming households benefiting from improvements in irrigation facilities, crop diversity, and agricultural productivity.⁸ Similarly, evaluations of the Accelerated Irrigation Benefits Program component describe the program as performing satisfactorily on parameters such as relevance, efficiency, impact, and equity. At the same time, the sustainability aspects of the program (including financial, operational, and environmental) have been graded as needing improvement.⁹

Figure 4: Area under rain-fed agriculture as a percentage of the net cultivated area as on the end of the current or previous FY (in per cent, FY 16-17)



Source: NITI Aayog, Composite Water Management Index 2.0, 2019

⁷ Ministry of Water Resources, River Development and Ganga Rejuvenation, Irrigation growth in India – Prospects, Initiatives, and Challenges, 2017

⁸ NITI Aayog, Evaluation Study On Command Area Development & Water Management Programme, 2015

⁹ PIB, 24.35 Lakh Additional Irrigation Potential Created Under PMKSY During 2016-2022, 2023; DMEO & NITI Aayog, Evaluation of Centrally Sponsored Schemes in Water Resources, Environment and Forest Sector – Approach and Methodology, 2020

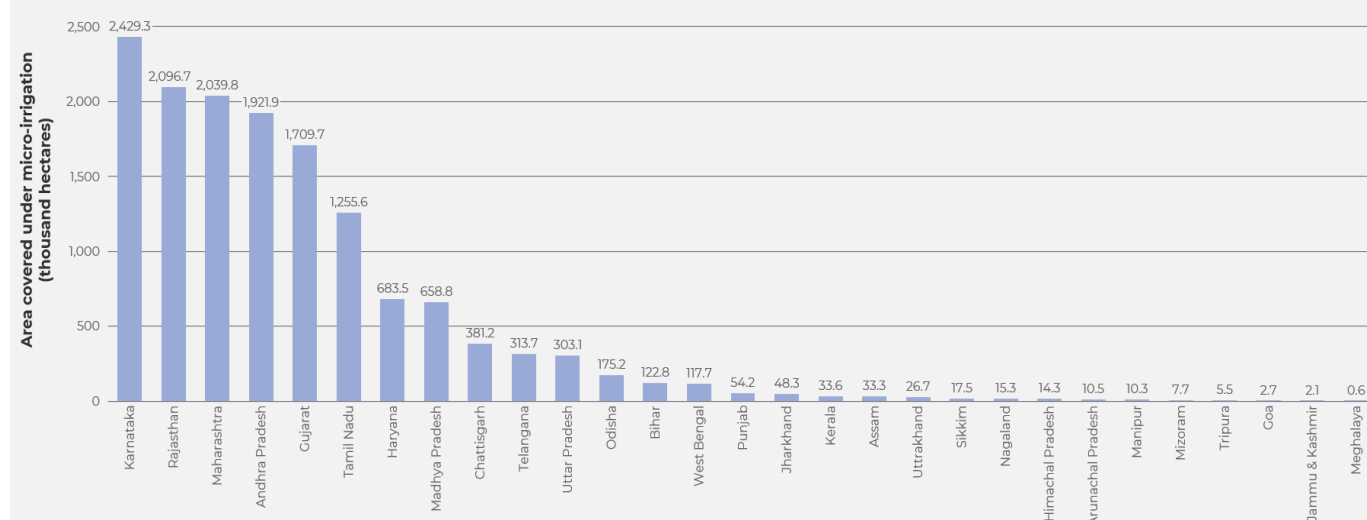
Water-Use Efficiency (WUE)

Water use efficiency in IWM involves maximizing crop yield while minimizing water wastage, and optimizing the use of available water resources for irrigation.

There is a need to improve WUE in agriculture:

The economic WUE in agriculture for India is 0.42 USD/m³ compared to 38.85 USD/m³ and 24.83 USD/m³ for the industrial and services sectors.¹⁰ Similarly, the economic WUE for agriculture in countries such as Israel (2.12 USD/ m³), China (2.1 USD), and Bangladesh (0.84 USD) is significantly more than in India. Put differently, India utilizes two to three times more water per unit of major food crop production compared to nations like China, Brazil, and the USA.¹¹ Planting crops like millets can significantly enhance water use efficiency in Indian food crops, yet their adoption remains limited due to the prevailing demand and supply side factors. Similarly, increasing the adoption of micro-irrigation across Indian agriculture can lead to significant improvements in WUE but progress needs to be carefully monitored to ensure swift uptake.

Figure 5: State-wise Area Covered under Micro Irrigation, 2022



Source: Ministry of Agriculture and Farmers Welfare, Agricultural Statistics at a Glance, 2022

¹⁰ UN Water, SDG 6 Progress, 2019

¹¹ United Nations, The Perfect storm: Pathways to manage India's water sector sustainably, 2018

Water Pricing

Water pricing in IWM involves implementing mechanisms to design and enforce appropriate price systems for agricultural water to reduce over-extraction of groundwater. This needs to be addressed in a balanced manner to ensure livelihoods and water sustainability with a special focus on smallholder farmers.

A majority of states in India charge for electricity used on farms for agriculture:

80% of states and UTs have started charging for electricity for tube wells and pumps used for agriculture.¹² However, a large fraction of the States that do report charging for electricity, use a flat rate method either fully or in some parts of the State. Only 6 out of the 27 reporting States & UTs have a fully metered approach to charging for electricity given to tube wells and pumps.¹³

Equity and Justice

Equity and justice in IWM involve equitable irrigation water distribution among all stakeholders, ensuring access and allocation that is both inclusive and ethical.

There is low irrigation water access for tail-end farmers:

Tail-end farmers located at the end of the canal system in the command area receive less water for irrigation due to overutilization by the farmers located closer to the water source.¹⁴ A regional study in Punjab and Haryana showed that 70% of the tail-end farmers got 54-70% less water than their entitled amount.¹⁵

Absence of irrigation systems results in groundwater extraction:

Due to the partial penetration of formal irrigation systems across the country, more than 50% of Indian agriculture is rainfed and relies on groundwater extraction. This drives severe groundwater depletion in states like Punjab, Haryana, Uttar Pradesh, and Maharashtra.¹⁶

There is a need for improving irrigation water access to marginalized groups:

Marginalised communities such as women, smallholder farmers, and tribal groups often lack or have less land ownership. Additionally, since water rights are tied with formal land rights in India, this leads to poor water availability for irrigation for these groups who don't have land titles.¹⁷

Governance and Capacity Building

IWM is a state subject, implying that the primary responsibility for policy formulation and implementation lies with states, with the Centre providing an overall governance framework, financial and technical support, and coordination. The Ministry of Jal Shakti is the nodal agency for IWM. Additionally, the Ministry of Agriculture and Farmers' Welfare works to improve water-use efficiency, crop diversification, and irrigation infrastructure in the sector to reduce water required for irrigation. At the state level, irrigation or water resource departments play a crucial role.¹⁸ At the local level, given the

¹² NITI Aayog, Composite Water Management Index 2.0, 2019

¹³ NITI Aayog, Composite Water Management Index 2.0, 2019

¹⁴ ICRISAT, Pradhan Mantri Krishi Sinchai Yojana: Enhancing the Impact through Demand Driven Innovations, 2016

¹⁵ ICRISAT, Pradhan Mantri Krishi Sinchai Yojana: Enhancing the Impact through Demand Driven Innovations, 2016; CEEW, Reallocating Water for India's Growth, 2020

¹⁶

¹⁷ Varsha Khandker, Vasant P. Gandhi, Nicky Johnson, Gender Perspective in Water Management: The Involvement of Women in Participatory Water Institutions of Eastern India, 2020

¹⁸ Ministry of Jal Shakti, Organizational history of the Department of Water Resources, River Development and Ganga Rejuvenation

challenges of social, class, and gender dynamics in IWM in India, institutions such as panchayats and WUAs are crucial for ensuring equity and justice in irrigation water distribution.¹⁹

Budgetary focus on IWM has been reduced for 2023-24:

While the budget for “Water” as a domain has increased to ~INR 1,12,478 crore, up 15% compared to last year; however, the budget provided to the Ministry of Agriculture and Farmers’ Welfare for Per Drop More Crop for the micro irrigation component has decreased by 25% from last year.²⁰

Budgetary focus for the promotion of less-water-intensive crops is also low:

Despite the increasing emphasis of the government on promoting less water-intensive crops like millet, the budget allowances are not aligned with the objective given the assured procurement, and MSP focus still favours rice and wheat. While the allocations for indigenous and agroecological-sensitive cropping patterns to promote less water use are dismal.²¹

¹⁹ ICRISAT, Pradhan Mantri Krishi Sinchai Yojana: Enhancing the Impact through Demand Driven Innovations, 2016; CEEW, Reallocating Water for India’s Growth, 2020; Varsha Khandker, Vasant P. Gandhi, Nicky Johnson, Gender Perspective in Water Management: The Involvement of Women in Participatory Water Institutions of Eastern India, 2020

²⁰ Ministry of Finance, Union Budget at a Glance - 2023-24, 2023

²¹ Ministry of Finance, Union Budget at a Glance - 2023-24, 2023; NABARD, Water Productivity Mapping of Major Indian Crops, 2018; PRS, MSP and Public Procurement, 2020; United Nations, The Perfect storm: Pathways to manage India’s water sector sustainably, 2018

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by adopting effective irrigation water management practices. These revolve primarily around the potential for improved irrigation water efficiency, greater agricultural productivity, and improved groundwater resilience.



Improved agricultural productivity, farmer incomes, and water savings through the promotion of less water-intensive crops and micro-irrigation:

Growing less water-intensive crops and using micro-irrigation can result in significant returns for the farmers while reducing water consumption. Some studies suggest that economic returns can be enhanced by 147% by shifting from staple crops such as paddy and wheat to commercially high-value crops such as pulses.²² According to the Ministry of Agriculture and Farmers Welfare, the returns to farmers for crops like pulses and oilseeds are expected to be 50-90%.²³ As staple crops such as wheat and paddy consume more water, this transition suggests that water-centric interventions will improve agricultural productivity and enhance farmers' incomes. Similarly, micro-irrigation water use efficiency is as high as 80-95%, compared to 30-50% in the case of conventional flood irrigation, and greater adoption can help conserve water resources.²⁴



Targeted focus in irrigation programs to improve food production outcomes:

Paddy, sugarcane, and wheat occupy about 40% of the gross cropped area of India but consume more than 80% of the irrigation water available.²⁵ States cultivating these crops, such as Punjab and Haryana, also have the lowest levels of irrigation water productivity. In contrast, states like Chhattisgarh and Jharkhand have higher irrigation water productivity for rice but low irrigation coverage (32% and 3%, respectively).²⁶ Food production in these states can be improved by bringing more area under irrigation, enhancing food production while reducing the average water consumed per unit of crop output in the country.



Leveraging WUAs to drive change management in irrigation practices:

WUAs will have a critical channel to improve irrigation water use efficiency in India. Behaviour change and training programs coupled with frequent hyperlocal data collection at the level of individual WUAs will be critical in ensuring the shift towards improved irrigation practices in Indian farming. Care must be taken, however, to ensure equity so that benefits accrue widely to smallholder farmers across all social groups.



Reviving effective traditional irrigation practices:

Reviving certain traditional irrigation practices in India can help ensure sustainable agriculture in the face of India's water crisis, for instance, reviving traditional water wells and enabling them to serve as water reservoirs.²⁷

Challenges

²² NABARD, ICRIER, Water Productivity Mapping of Major Indian Crops, 2018

²³ Ministry of Agriculture and Farmers Welfare, Annual Report 2020-21, 2021

²⁴ Ministry of Agriculture & Farmers Welfare, Adoption of Methods of Irrigation, 2019

²⁵ NABARD, Water Productivity Mapping of Major Indian Crops, 2018

²⁶ NABARD, Water Productivity Mapping of Major Indian Crops, 2018

²⁷ UN, CEEW, The Perfect Storm – Pathways to Managing India's Water Sector Sustainably, 2018

This section highlights some of the major challenges and constraints facing irrigation water management in India.



Underutilised irrigation potential, policy focus on water-guzzling crops, and low adoption of micro-irrigation:

Only 19% of the net irrigated area in India is cultivated through micro-irrigation systems due to a lack of awareness, high upfront costs, insufficient affordable financing mechanisms, and challenges with operations & maintenance.²⁸ Further, there is also a heavy reliance on water-intensive crops. Although the Government of India has announced MSP for 23 crops, still ~80% of the crops procured under the MSP regime in 2019-20 were rice and wheat – which are highly water-intensive.²⁹



The potential of WUAs and PIM not fully realized:

WUA members often lack awareness of various responsibilities, such as the preparation of water availability maps, crop plans, financial planning, and conflict resolution mechanisms leading to poor operations and PIM.³⁰ PIM in India also suffers from inadequate representation and participation of women, small landholding farmers, and farmers belonging to marginalised tribal and caste groups.³¹

²⁸ Ministry of Agriculture and Farmers Welfare, Agricultural Statistics at a Glance, 2021; TERI, Water and Agriculture in India, 2017

²⁹ NABARD, Water Productivity Mapping of Major Indian Crops, 2018; PRS, MSP and Public Procurement, 2020; United Nations & Council On Energy, Environment, And Water (CEEW), The Perfect storm: Pathways to manage India's water sector sustainably, 2018

³⁰ IRMA, Institution and Socio-economic Assessment of Water Users Association under WBADMIP, 2020

³¹ Khandker et al (2020), Gender perspective in water management: The involvement of women in participatory water institutions of eastern India', Water, 12, 196

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee irrigation water management.



Ministry of Jal Shakti

Oversees the management and conservation of water resources in India, ensuring equitable access to safe and clean water while promoting sustainable water practices such as irrigation.



Ministry of Agriculture & Farmers' Welfare

Formulates and implements policies and programs to enhance irrigation infrastructure, promote efficient water use in agriculture, develop rainfed areas, and support farmers in adopting sustainable water management practices for improved agricultural productivity.



Central Ground Water Board

National agency responsible for providing scientific and technical inputs for management, exploration, monitoring, assessment, recharge, and regulation of groundwater resources of the country, critical for mapping groundwater requirements for irrigation.



Central Groundwater Authority

Set up in 1997 under the Environment (Protection) Act 1986 for regulation, conservation, protection of groundwater and mass awareness; wide powers including the power to impose a penalty on any person, company, or government department.



National Water Development Agency

Carries out water balance and scientific studies for optimum utilisation of water resources of the Peninsular River system, including water use for irrigation.



National Water Informatics Centre

Maintains a central repository of water resources in the country.

Major policies and programs

This section highlights existing policies related to irrigation water management and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



Pradhan Mantri Krishi Sinchayee Yojana, 2015 (PMKSY)³²

PMKSY aims to enhance irrigation coverage and improve water use through a targeted approach, providing comprehensive solutions for source creation, distribution, management, field application, and extension activities. Some key features involve expanding the cultivable land area with reliable access to irrigation, improving the efficiency of water usage on farms and promoting the adoption of precision irrigation and other water-saving technologies among others. The Command Area Development & Water Management (CADWM), which aimed to bridge the gap between irrigation potential created and utilised was also brought under this scheme in 2015.



Accelerated Irrigation Benefit Programme (AIBP)³³

AIBP aims to provide Central Assistance to States for large and medium irrigation projects nationwide, aiming to expedite the completion of progressed irrigation projects hindered by financial limitations.

³² Ministry of Agriculture & Farmers' Welfare Government of India, Operational guidelines of Pradhan Mantri Krishi Sinchayee Yojana (PMKSY), 2015
³³ Ministry of Jal Shakti, Assistance to States Under AIBP, 2021

Way forward

This section highlights considerations for policymakers seeking to advance policy formulation and implementation in this sector.

Irrigation water management is going to be a critical determinant of India's future food security and farmer prosperity while also ensuring the sustainability of water resources and environments. It is important to recognise that efficient irrigation water use involves not just technical fixes but also shaping the behaviours and decisions of millions of farmers embedded in their localised contexts.

Some themes that would enable progress include:

 Enhancing data systems and analytical capabilities	- Invest in improved data systems and analytics related to irrigation water use, efficiency benchmarks, impacts, and outcomes to: - diagnose irrigation challenges, - evaluate initiatives like Water User Associations, and - design/inform effective water management policies such as payment for ecosystem services, based on insights from irrigation practices and outcomes.
 Transitioning to less-water intensive cropping	- Design state-level "just transition" roadmaps to promote less water-intensive cropping practices such as - aligning farmer incentives through assured demand and minimum support prices for crops like millets - providing extension services to educate farmers on agronomic practices that support the shift away from water-intensive crops in specific regions.
 Fostering public-private partnerships for micro-irrigation	- Set up collaborations between public and private sectors to scale up micro-irrigation coverage by - providing public subsidies to incentivise private sector involvement in micro-irrigation initiatives and - exploring innovative, outcome-based finance models to increase penetration of micro-irrigation products among small and marginal farmers.
 Empowering communities	Collaborate with agricultural extension partners, civil society organisations, and grassroots media to disseminate hyperlocal and interactive information on irrigation best practices as part of behaviour change programs and awareness campaigns.

through the dissemination of information	
 Supporting community-led water management	• Support civil society initiatives focused on participatory groundwater monitoring, aquifer mapping, and community water budget-linked agriculture plans for scenario planning and trade-off analysis to equip communities with knowledge and tools to make informed decisions about irrigation water use.

Recommended readings

Ministry of Jal Shakti. (2023). *Standing Committee on Water Resources (2022-23) Groundwater: A Valuable But Diminishing Resource*.

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